

# Spatio-Temporal Forecasting System for Hourly Weather Data

## Integrating Database, Kriging, and Machine Learning

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Spatial Statistics Final Project

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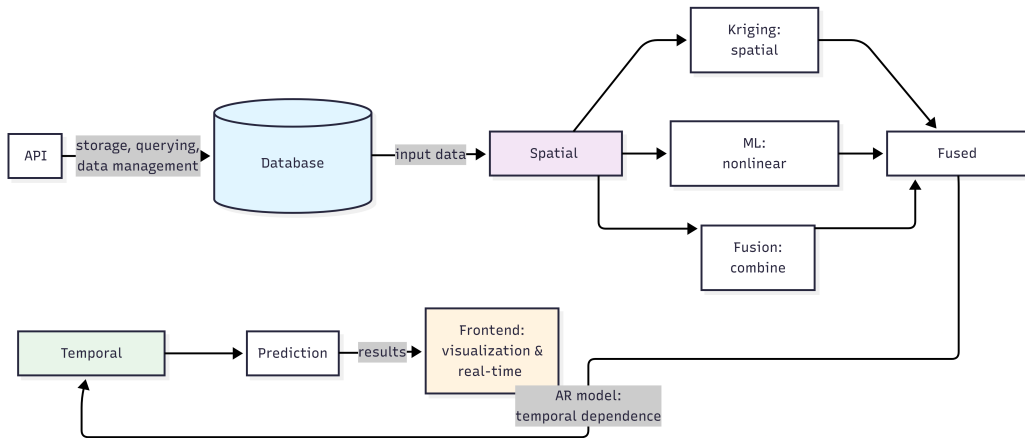
# Motivation

- Weather station data exhibit:
  - Temporal dependence
  - Spatial dependence
  - Non-stationarity
- Time series models ignore spatial structure
- Kriging assumes linearity and stationarity
- Machine learning ignores spatial dependence

## Goal

Develop an end-to-end spatio-temporal forecasting system

# System Workflow



# Step 1: Data Ingestion

- Source: Central Weather Administration API
- Frequency: hourly update
- Raw data stored in:

`observations_hourly`

- Station metadata stored in:

`stations`

# Step 1: data resource

Swagger  
1.6.0

Select a definition openAPI

## 中央氣象署開放資料平臺之資料擷取API 1.0.0 OAS 2.0

[ Base URL: /api ]

提供目前開放資料之擷取API

Schemes  
HTTPS

### 觀測

GET /v1/rest/datastore/O-A0001-001 氣象觀測站-全測站即時氣象資料

氣象觀測站-全測站即時氣象資料

Try it out

| Name                                         | Description                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Authorization</b> <small>required</small> | 氣象開放資料平台會員授權碼                                                                                                                                                                         |
| string (query)                               | Authorization                                                                                                                                                                         |
| <b>limit</b>                                 | 限制最多回傳的資料，預設為回傳全部筆數                                                                                                                                                                   |
| number (int) (query)                         | limit                                                                                                                                                                                 |
| <b>offset</b>                                | 指定從第幾筆後開始回傳，預設為第 0 筆開始回傳                                                                                                                                                              |
| number (int) (query)                         | offset                                                                                                                                                                                |
| <b>format</b>                                | 回傳資料格式，預設為 json 格式                                                                                                                                                                    |
| string (query)                               | Available values: JSON, XML                                                                                                                                                           |
|                                              | --                                                                                                                                                                                    |
| <b>StationId</b>                             | 測站站號，請參考 <a href="https://hdp.cwb.gov.tw/static/state.html">https://hdp.cwb.gov.tw/static/state.html</a> ，可輸入多個站號，大小寫必須完全符合，預設為全部回傳                                                   |
| array of strings (query)                     |                                                                                                                                                                                       |
| <b>StationName</b>                           | 測站站名，請參考 <a href="https://hdp.cwb.gov.tw/static/state.html">https://hdp.cwb.gov.tw/static/state.html</a> ，可輸入多個站名，預設為全部回傳，若使用「StationId」，則參數「StationName」的篩選資料將會失效，只會回傳StationId符合的資料 |
| array of strings (query)                     |                                                                                                                                                                                       |

# Step 1:data of all station

|  | id  | station_id | station_name | datetime            | lat       | lon        | temperature | humidity | wind_speed | wind_direction | pressure |
|--|-----|------------|--------------|---------------------|-----------|------------|-------------|----------|------------|----------------|----------|
|  | 822 | CAP040     | 國一N361K      | 2026-04-13 10:00:00 | 22.680555 | 120.328611 | 29.6        | 72       | 1.8        | 200            | 1011.2   |
|  | 823 | CAP060     | 國三S385K      | 2026-04-13 10:00:00 | 22.767777 | 120.444444 | 28.8        | 62       | 1.2        | 80             | 1003.2   |
|  | 824 | CAP070     | 高雄燈塔         | 2026-04-13 10:00:00 | 22.614594 | 120.26468  | 27.5        | 75       | 1.8        | 230            | 1011.1   |
|  | 825 | CAQ010     | 國三N425K      | 2026-04-13 10:00:00 | 22.480277 | 120.531666 | 28.3        | 70       | 1.5        | 220            | 1010.5   |
|  | 826 | CAQ020     | 國三S415K      | 2026-04-13 10:00:00 | 22.568    | 120.528    | 28.7        | 64       | 1.3        | 280            | 1011     |
|  | 827 | CAQ030     | 崎峰國小         | 2026-04-13 10:00:00 | 22.42565  | 120.49521  | NULL        | NULL     | 2.6        | 240            | 1014.1   |
|  | 828 | CAUB30     | 國五S031K      | 2026-04-13 10:00:00 | 24.819144 | 121.782378 | 28          | 78       | 2.4        | 50             | 1012.1   |
|  | 829 | CAUB40     | 國五S041K      | 2026-04-13 10:00:00 | 24.734764 | 121.781922 | 28.1        | 76       | 1.3        | 90             | 1010.1   |
|  | 830 | CAUB50     | 國五S047K      | 2026-04-13 10:00:00 | 24.675456 | 121.794583 | 27.7        | 75       | 2.2        | 60             | 1010.1   |
|  | 831 | CAV270     | 國一S336K      | 2026-04-13 10:00:00 | 22.894167 | 120.271178 | 28.6        | 72       | 2.4        | 200            | 1010.9   |
|  | 832 | CAV280     | 國三S378K      | 2026-04-13 10:00:00 | 22.825958 | 120.415519 | 28.3        | 75       | 1.5        | 280            | 1001.2   |
|  | 833 | CAIW010    | 東引           | 2026-04-13 10:00:00 | 26.358836 | 120.483631 | 19.1        | 98       | 1.6        | 120            | 1007.8   |
|  | 834 | E2H360     | 鹽埕池          | 2026-04-13 10:00:00 | 23.918333 | 120.885278 | 27.3        | 53       | 1.9        | 130            | 940.7    |
|  | 835 | E2HA20     | 畢祿溪站         | 2026-04-13 10:00:00 | 24.226944 | 121.308522 | 20.5        | 17       | 0.7        | 60             | 790      |
|  | 836 | E2K600     | 四湖植物園        | 2026-04-13 10:00:00 | 23.676986 | 120.163433 | 29.3        | 86       | 1.6        | 240            | 1014.1   |
|  | 837 | E2P980     | 六龜林試         | 2026-04-13 10:00:00 | 22.991703 | 120.645722 | 28.8        | 53       | 0.9        | 180            | 984.5    |
|  | 838 | E2P990     | 扇平林試         | 2026-04-13 10:00:00 | 22.9677   | 120.685703 | NULL        | NULL     | NULL       | NULL           | NULL     |
|  | 839 | E2S960     | 太麻里(2)       | 2026-04-13 10:00:00 | 22.598    | 120.99775  | 30.6        | 56       | 2.2        | 170            | 1012     |
|  | 840 | E2S980     | 太麻里(1)       | 2026-04-13 10:00:00 | 22.598922 | 120.980044 | 30.1        | 56       | 0.9        | 120            | 1003.7   |
|  | 841 | G2F820     | 農業試驗所        | 2026-04-13 10:00:00 | 24.031261 | 120.688133 | 28.9        | 67       | 3.1        | 260            | 1005.3   |
|  | 842 | G2L020     | 農試嘉義分所       | 2026-04-13 10:00:00 | 23.485028 | 120.473806 | 29.4        | 62       | 2.4        | 250            | 1005.9   |
|  | 843 | G2M350     | 農試溪口農場       | 2026-04-13 10:00:00 | 23.58565  | 120.408111 | 27.9        | 70       | 2.9        | 230            | 1012.3   |
|  | 844 | G2P820     | 農試鳳山分所       | 2026-04-13 10:00:00 | 22.646611 | 120.356083 | 29.9        | 57       | 1.1        | 150            | 1011.3   |
|  | 845 | K2E360     | 苗栗農改         | 2026-04-13 10:00:00 | 24.495722 | 120.829111 | 30.2        | 53       | 2.6        | 260            | 1000.6   |
|  | 846 | K2E710     | 苗改生物防...     | 2026-04-13 10:00:00 | 24.422861 | 120.871833 | 29.2        | 49       | 1.8        | 190            | 981.3    |
|  | 847 | K2F750     | 植苗繁殖         | 2026-04-13 10:00:00 | 24.225972 | 120.800917 | 26.2        | 62       | 2.3        | 240            | 961.8    |
|  | 848 | U2H480     | 臺大溪頭         | 2026-04-13 10:00:00 | 23.670306 | 120.798194 | 23.3        | 64       | 0.9        | 320            | 887.3    |
|  | 849 | U2HA30     | 臺大和社         | 2026-04-13 10:00:00 | 23.590856 | 120.888864 | 27.4        | 58       | 1          | 80             | 927.3    |
|  | 850 | U2HA40     | 臺大內茅埔        | 2026-04-13 10:00:00 | 23.689722 | 120.851117 | 29.1        | 45       | 0.6        | 30             | 956.8    |
|  | 851 | U2HA50     | 臺大竹山         | 2026-04-13 10:00:00 | 23.756972 | 120.681556 | NULL        | NULL     | NULL       | NULL           | NULL     |
|  | 852 | V2K610     | 大庄合作社        | 2026-04-13 10:00:00 | 23.8068   | 120.367939 | 27.9        | 74       | 2.3        | 230            | 1012.4   |
|  | 853 | V2K620     | 麥寮合作社        | 2026-04-13 10:00:00 | 23.753836 | 120.294783 | 28.2        | 66       | 3.1        | 230            | NULL     |

# Step 1: data of one station in different time

|   | id    | station_id | station_name | datetime            | lat       | lon        | temperature | humidity | wind_speed | wind_direction | pressure |
|---|-------|------------|--------------|---------------------|-----------|------------|-------------|----------|------------|----------------|----------|
| ▶ | 1     | C0TB40     | 崇德           | 2026-04-13 10:00:00 | 24.166144 | 121.657414 | 29.1        | 66       | 1.6        | 32             | 1011.3   |
|   | 854   | C0TB40     | 崇德           | 2026-04-13 11:00:00 | 24.166144 | 121.657414 | 30.3        | 64       | 1.4        | 47             | 1010.7   |
|   | 1707  | C0TB40     | 崇德           | 2026-04-13 12:00:00 | 24.166144 | 121.657414 | 29.2        | 67       | 2.1        | 74             | 1010.1   |
|   | 2560  | C0TB40     | 崇德           | 2026-04-13 13:00:00 | 24.166144 | 121.657414 | 28.1        | 73       | 2.4        | 115            | 1009.5   |
|   | 3413  | C0TB40     | 崇德           | 2026-04-13 14:00:00 | 24.166144 | 121.657414 | 27.9        | 70       | 3.4        | 134            | 1008.8   |
|   | 4266  | C0TB40     | 崇德           | 2026-04-13 15:00:00 | 24.166144 | 121.657414 | 27.6        | 72       | 2.2        | 128            | 1008.4   |
|   | 5119  | C0TB40     | 崇德           | 2026-04-13 16:00:00 | 24.166144 | 121.657414 | 27.4        | 72       | 2.3        | 130            | 1008.4   |
|   | 5972  | C0TB40     | 崇德           | 2026-04-13 17:00:00 | 24.166144 | 121.657414 | 27.6        | 72       | 1          | 155            | 1008.4   |
|   | 6825  | C0TB40     | 崇德           | 2026-04-13 18:00:00 | 24.166144 | 121.657414 | 26.6        | 75       | 0.7        | 272            | 1008.9   |
|   | 7678  | C0TB40     | 崇德           | 2026-04-13 19:00:00 | 24.166144 | 121.657414 | 26.1        | 76       | 0.6        | 261            | 1009.5   |
|   | 8531  | C0TB40     | 崇德           | 2026-04-13 20:00:00 | 24.166144 | 121.657414 | 25.5        | 78       | 1          | 248            | 1010.2   |
|   | 9384  | C0TB40     | 崇德           | 2026-04-13 21:00:00 | 24.166144 | 121.657414 | 24.6        | 80       | 1.2        | 242            | 1010.7   |
|   | 10237 | C0TB40     | 崇德           | 2026-04-13 22:00:00 | 24.166144 | 121.657414 | 24.4        | 79       | 0.9        | 289            | 1010.8   |

## Step 2: Spatial Modeling

### Kriging

- Uses spatial coordinates
- Based on variogram
- Provides spatially consistent prediction

### Machine Learning

- Uses spatial coordinates  $(x, y)$
- Learns nonlinear spatial relationships
- Implemented via ensemble (Super Learner)

### Key Idea

Kriging models spatial structure, ML captures nonlinear spatial patterns

## Step 3: Model Fusion

$$\hat{Z}_{fusion}(u, t) = w(u)\hat{Z}_{ML}(u, t) + (1 - w(u))\hat{Z}_{KR}(u, t)$$

Where

$$w(u) = (\sigma_{KR}^2)^b, \quad b = b_0 + b_1 Z(u)$$

- $\hat{Z}_{ML}(u, t)$ : ML prediction
- $\hat{Z}_{KR}(u, t)$ : Kriging prediction
- $w(u)$ : weight based on kriging variance
- Because  $\sigma^2$  is standardize ,the interval between  $w(u)$  is  $[0, 1]$

## Step 3: Model Fusion

Property :

- If the value near the station, then Kriging model has the main influence.
- If the value far then station, then machine learning help us to predict the value.
- In complex situation like non-stational, trend, skew etc, mix model is better than only kriging.
- In simple situation like Gaussian, Kriging is little better than mix.

## Step 4: Temporal Modeling (AR)

- Apply autoregressive (AR) model at each location

- Input:

$$\hat{Z}_{fusion}(x, y, t - 1), \hat{Z}_{fusion}(x, y, t - 2), \dots$$

- Output:

$$\hat{Z}_{final}(x, y, t)$$

## Step 5: Prediction Storage

Results are stored in:

`predictions_hourly`

- Kriging prediction
- ML prediction
- Fusion prediction
- AR-based final prediction

### Database Role

Data storage, prediction tracking, and frontend query support

## Combined Machine Learning and Kriging Approach

- Reference:
  - Erten, Yavuz, and Deutsch (2023)
  - *Estimation with a Combined Machine Learning and Kriging Spatial Estimation*

### Machine Learning

- Captures nonlinear relationships
- Handles non-Gaussian data

### Kriging

- Models spatial dependence
- Provides unbiased interpolation

### Temporal Model (AR)

- Captures temporal dependence
- Models time series dynamics

# Expected Results

- Real-time spatio-temporal forecasting pipeline
- Comparison:
  - Kriging
  - Machine Learning
  - Combined spatial model
- Outputs:
  - Spatial heatmap
  - Temporal forecasts (time series)
  - Error analysis

# Conclusion

- Spatial modeling is achieved via Kriging and ML
- Temporal dependence is modeled separately using AR
- The proposed system separates spatial and temporal structures
- Fusion model leverages strengths of both spatial and temporal modeling

# To do list

- Compare with the weight where  $w(u) = 0, 1/2, 1$
- Think that how can i design with the interval  $b_0, b_1$
- Test which stardardize of the  $\sigma_{KR}^2(u)$  can be use (reference doesn't talk to this)